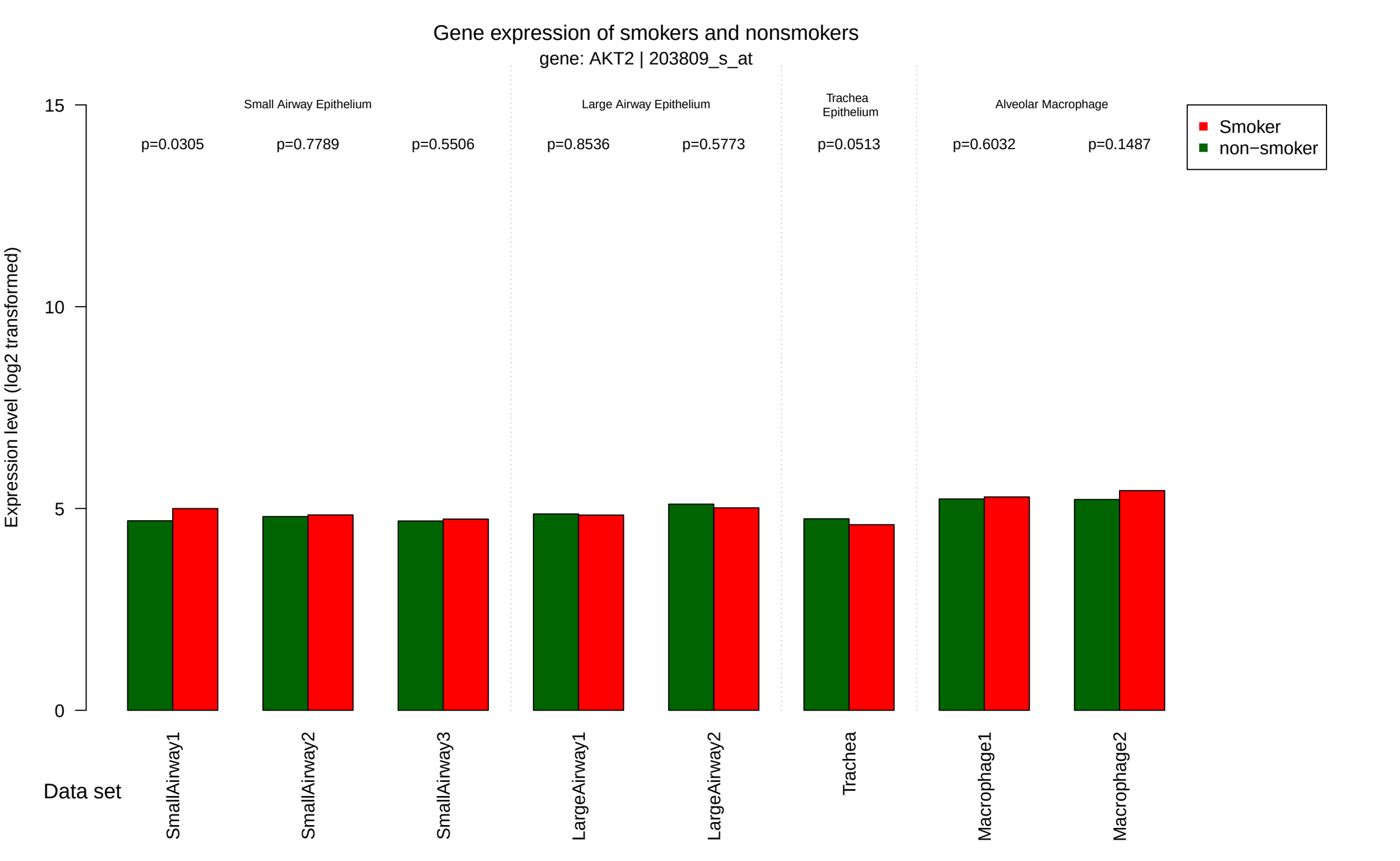
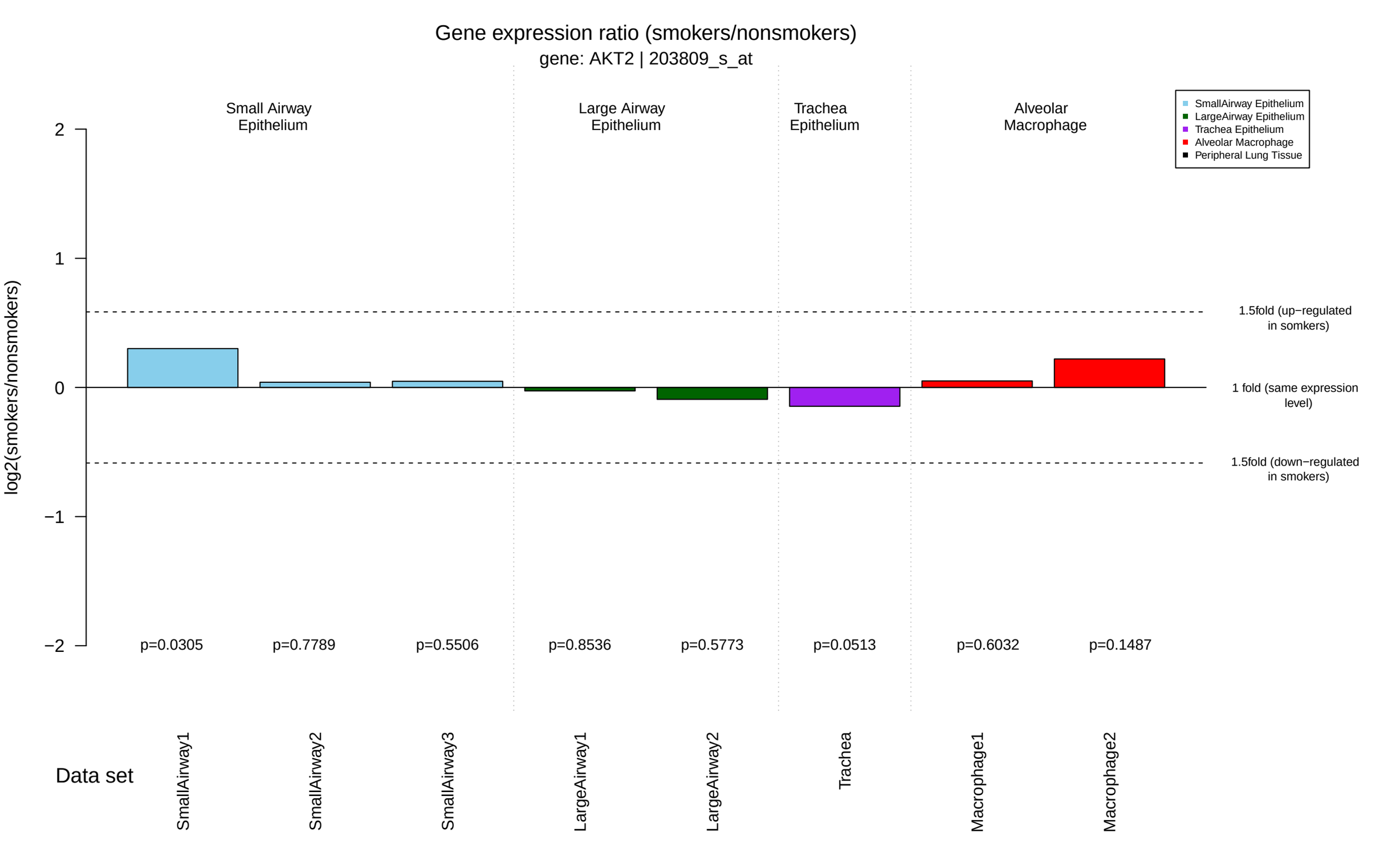
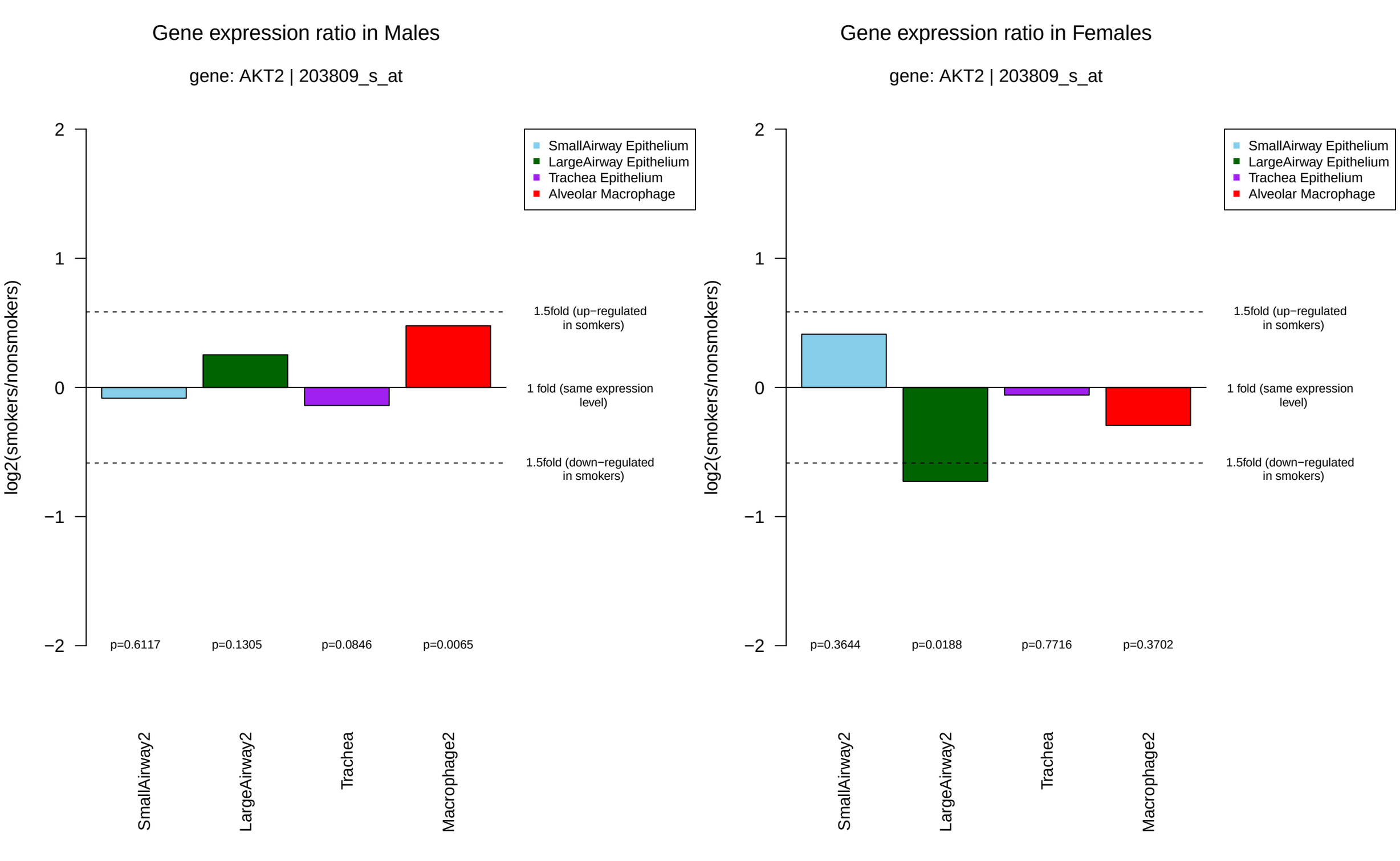
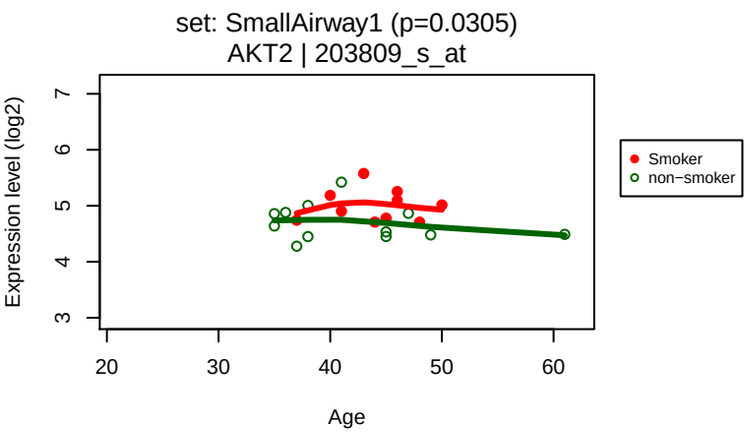
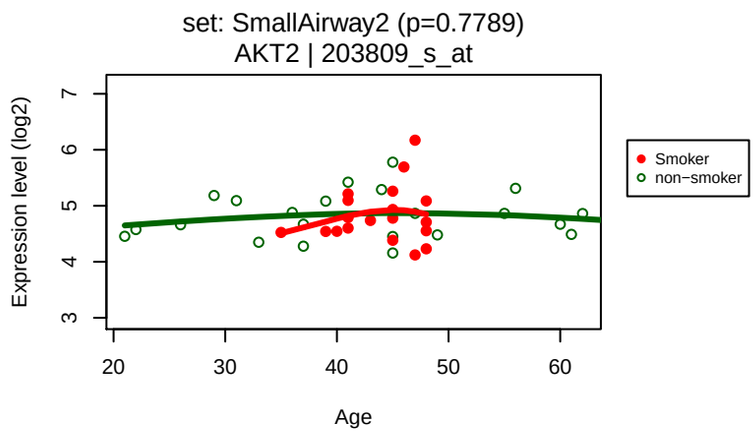
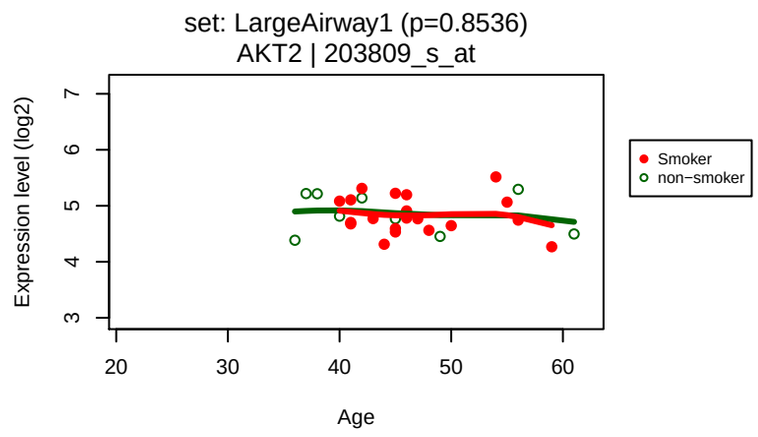

Gene expression of smokers and nonsmokers
gene: AKT2 | 203809_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0305
p=0.7789
p=0.5506
p=0.8536
p=0.5773
p=0.0513
p=0.6032
p=0.1487
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

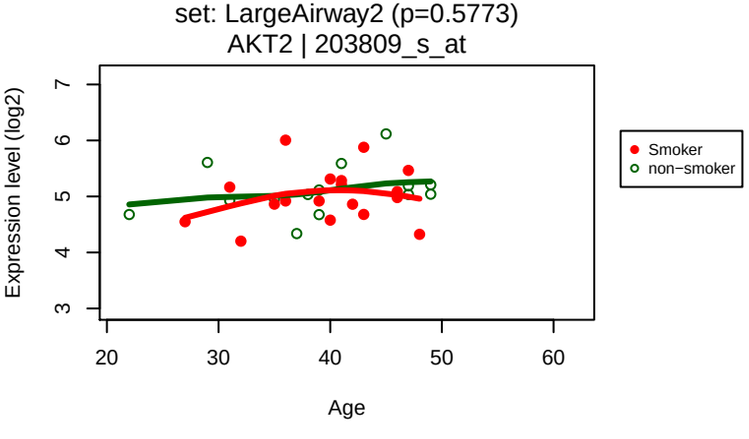

Gene expression ratio (smokers/nonsmokers)
gene: AKT2 | 203809_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0305
p=0.7789
p=0.5506
p=0.8536
p=0.5773
p=0.0513
p=0.6032
p=0.1487
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

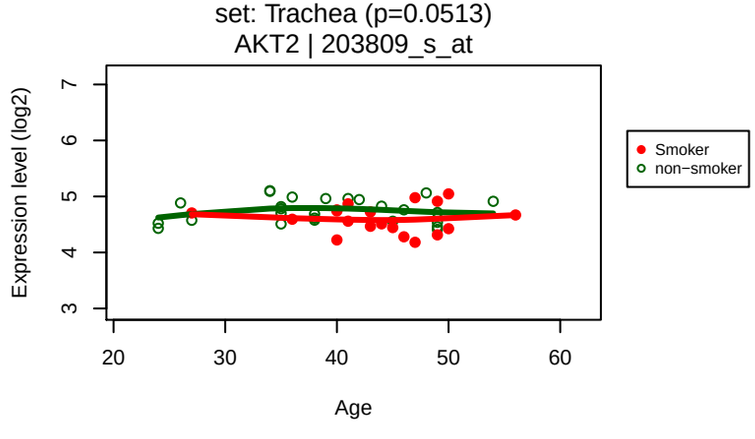

Gene expression ratio in Males
gene: AKT2 | 203809_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6117
p=0.1305
p=0.0846
p=0.0065
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT2 | 203809_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3644
p=0.0188
p=0.7716
p=0.3702
SmallAirway2
LargeAirway2
Trachea
Macrophage2

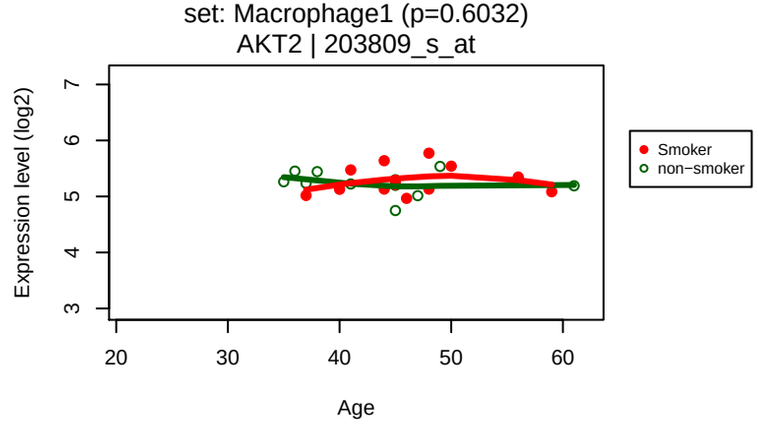

set: SmallAirway1 (p=0.0305)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker

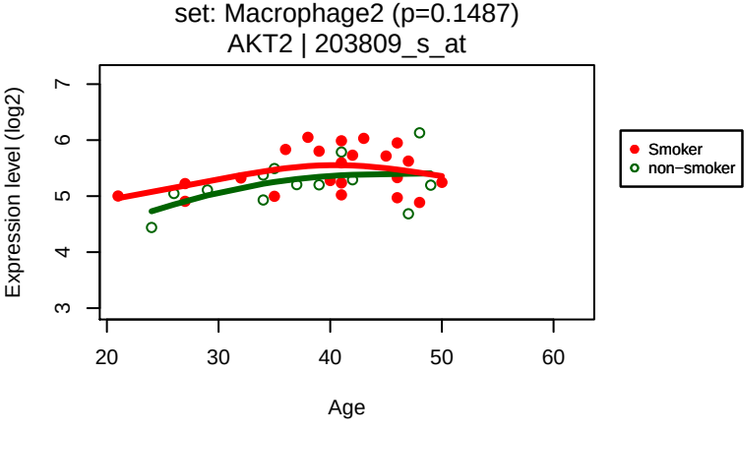

set: SmallAirway2 (p=0.7789)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker

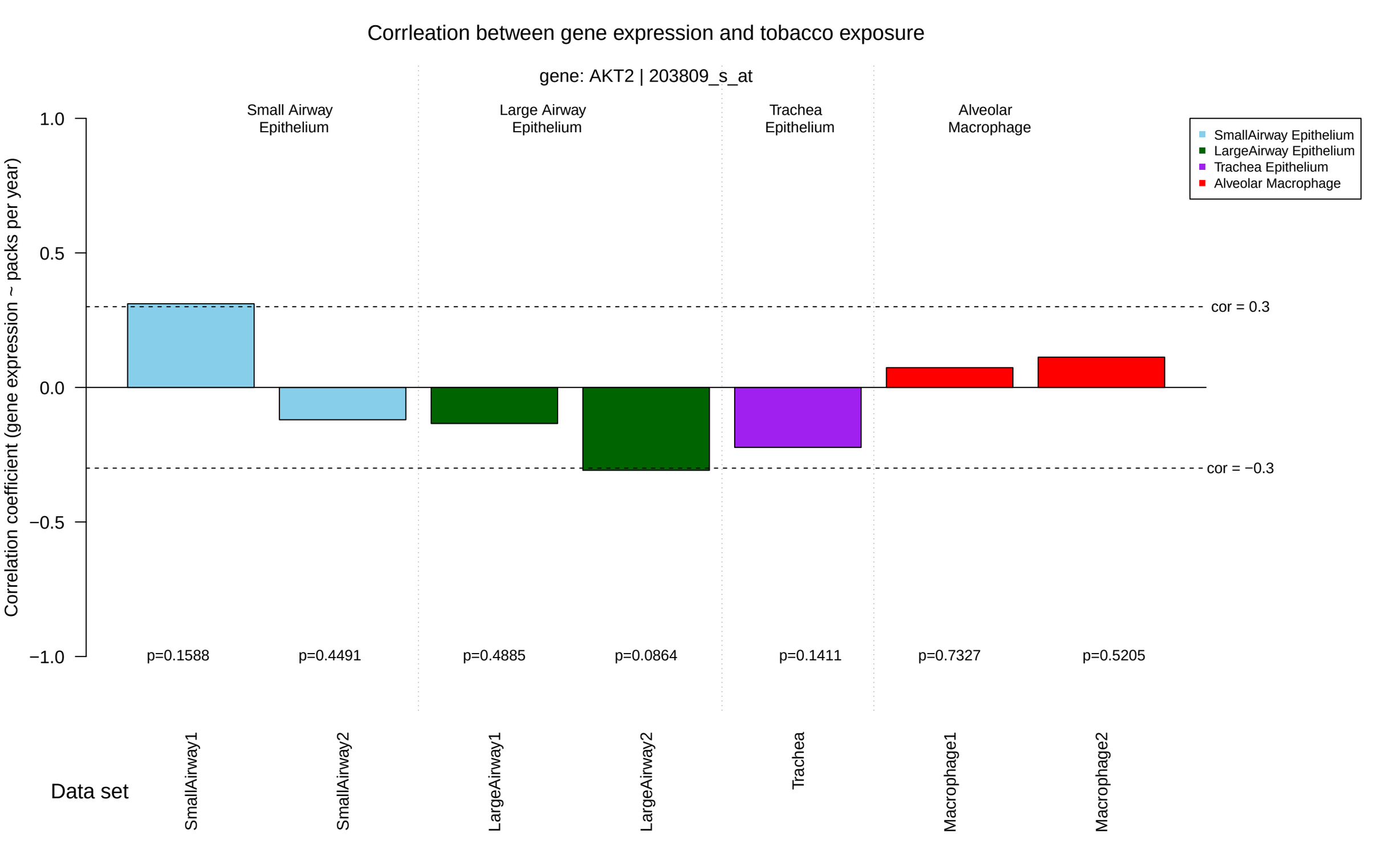

set: LargeAirway1 (p=0.8536)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5773)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0513)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6032)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1487)
AKT2 | 203809_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT2 | 203809_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1588
p=0.4491
p=0.4885
p=0.0864
p=0.1411
p=0.7327
p=0.5205
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2